This procedure is applicable to the following models:

Machine Model	**Server Option**		
	S1	**S2**	**S3**
V810 / V810i STD		√	√
V810 / V810i XXL		√	√
V810 / V810i S2EX		√	√
V810 / V810i Mini			
V810 / V810i XLT / XLTM / XLL		√	√
V810 / V810i XLW			
V810 / V810i S3		√	√

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	30 Apr 2024	First Release	Yeap Leong Wei
01	30 Oct 2024	Include STD & XXL for applicable model + content update	Yeap Leong Wei

Content

Scope

This procedure provides guidelines for the alignment of the XY axes Quantic Encoder Read Head.

Tool List

No	Part No	Description	Qty	UOM	Images
1	-	Ball Point Metric Allen Wrench Set	1	SET	
2	-	Metric T-Handle Wrench (M5 size)	1	PC	
3	-	Camera Tool	1	PC	
4	-	Tamper Proof Screw Driver Set	1	Set	
5	-	Cutter	1	PC	

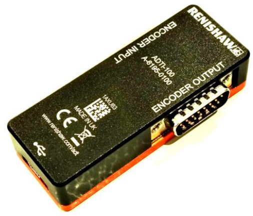
6	4XASPP-A440	Renishaw Quantic Encoder Alignment Kit	1	Set	 +  + 

Table 1: Tool List

X-Axis Encoder Head Alignment Procedure

1. **Read Head Spacer Preparation:** Obtain a read head spacer, ensuring it matches the required specifications outlined by the manufacturer for the alignment process.

   Note: the spacer was attached with the carton box of spare part 2675-0053

Figure 1: Read Head Encoder Spacer

2. **X-Axis Read Head Encoder Bracket Loosening:** Using the appropriate tool, carefully loosen the two screws located at the X-Axis read head encoder bracket. Take care not to over-loosen the screws to prevent any accidental detachment of components.

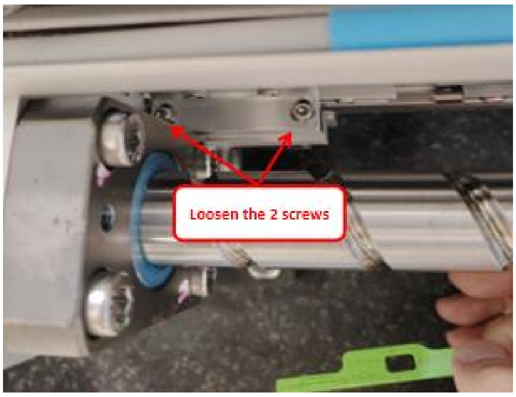


Figure 2: X-Axis Read Head Encoder

3. **Read Head Alignment Spacer Placement:** Place the read head alignment spacer between the read head encoder and the Quantic scale, ensuring it is positioned centrally and evenly aligned.

Figure 3: X-Axis Read Head Encoder with Spacer

4. **Read Head Encoder Adjustment:** Gently push the Read Head Encoder towards the read head spacer until it makes contact. Ensure that the spacer is fully engaged and seated properly. Once in position, tighten the two screws at the X-Axis read head encoder bracket securely. Apply torque evenly to prevent any misalignment or slippage during operation. Remove the spacer after securing the encoder bracket.

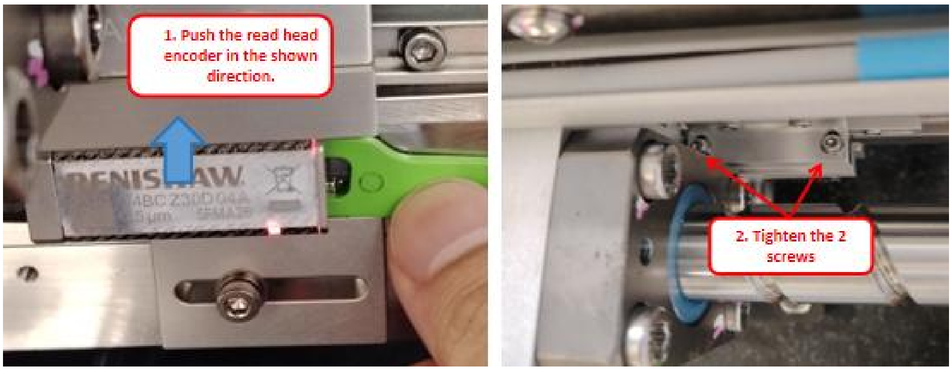


Figure 4: X-Axis Read Head Encoder Alignment

5. **Stage Movement Verification:** Manually push the stage from the minimum hard stop (located at the front access door) to the maximum hard stop (located at the rear access door). Throughout the movement, observe the rear head encoder LED, which should remain blinking green continuously. This indicates proper alignment and consistent signal detection as the stage moves along the axis.
Note: If the read head encoder LED turns RED/ORANGE at any location of the stage, it indicates potential misalignment. In such cases, route back to Step 1 for read head encoder alignment.
6. **SSCA Power Cycle:** Power off the SSCA (Servo System Control Amplifier) to reset the system.
7. **LED Status Change:** After powering on the SSCA, observe the change in status of the LED on the read head encoder. The blinking green LED will transition to blinking blue, indicating a change in operational mode.

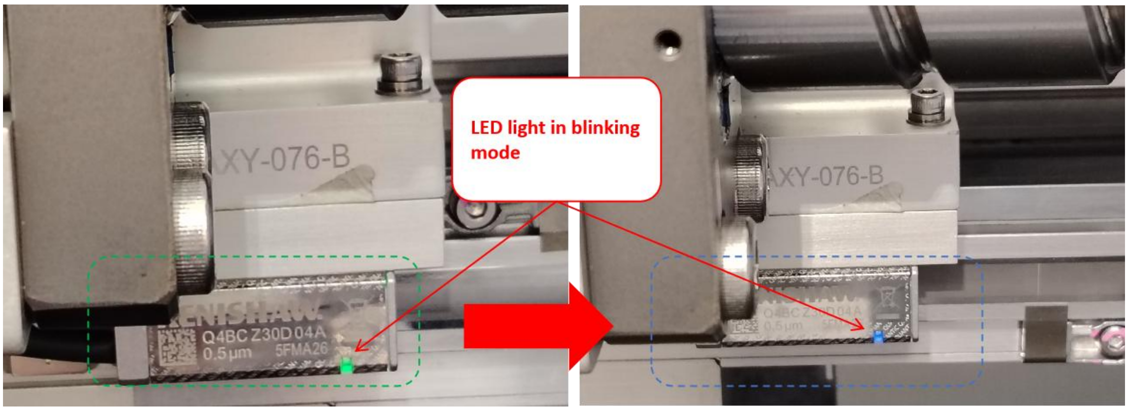


Figure 5: LED Status Change (Blinking Green to Blinking Blue)

8. **X Stroke Verification:** Using controlled force, push and pull the X stroke to execute movement along the axis. **Ensure that the read head encoder passes through the home sensor magnet** once per movement for both forward and backward directions. As the alignment progresses, the blinking blue LED will stabilize and appear as a static blue, indicating successful alignment completion.

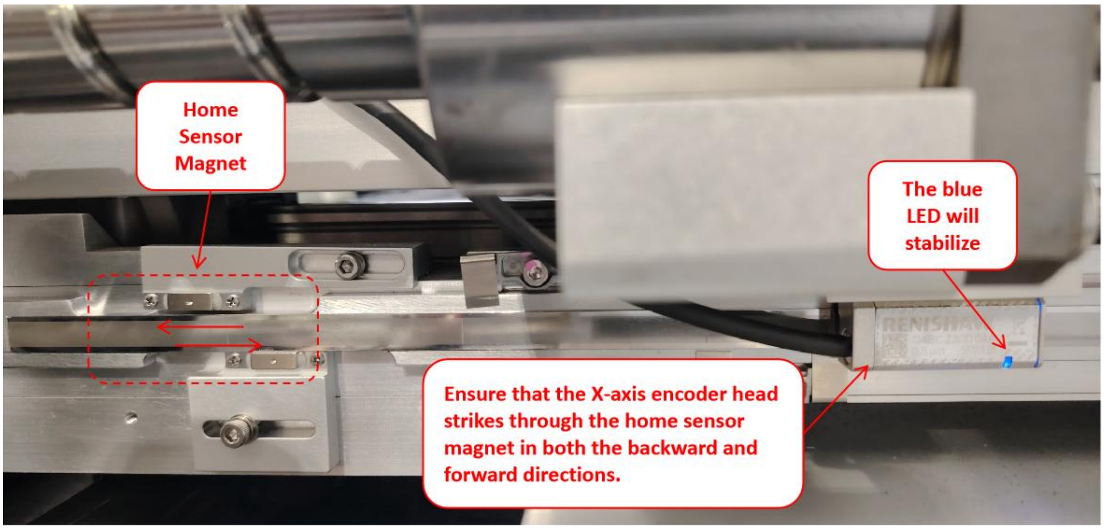


Figure 6: To Stabilize the blue LED for Read Head Encoder

9. **Block Read Head Encoder Sensor**: Use plastic bag to block the encoder head sensors for signal reset before proceed to adjustment and confirmation procedure.

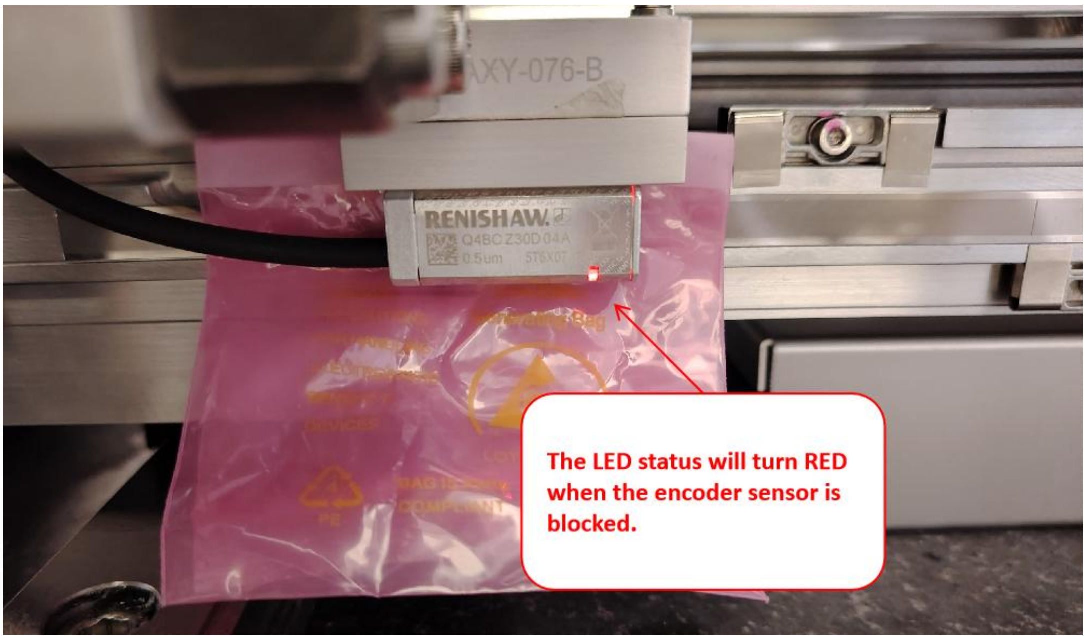


Figure 7: To Reset the LED Status for Read Head Encoder into Detection Mode

X-Axis Encoder Head Adjustment and Confirmation

10. **Connection Establishment:** Establish connections for the Advanced Diagnostic Tool (ADT), main server, encoder, and Kollmorgen kit according to the manufacturer's instructions.

    Note: Be ready with 4XASPP-A440 (Renishaw Quantic Encoder Alignment Kit)

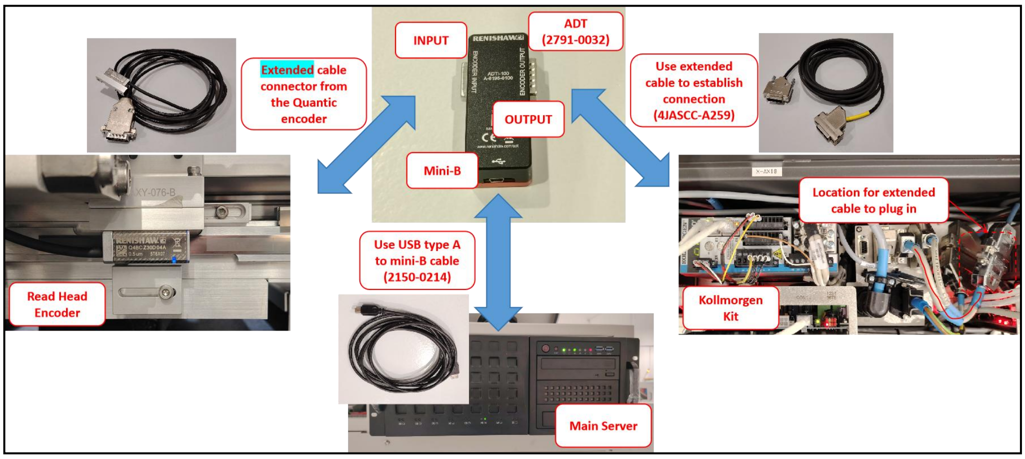


Figure 8: ADT, Main Server, Encoder and Kollmorgen Kit Connections

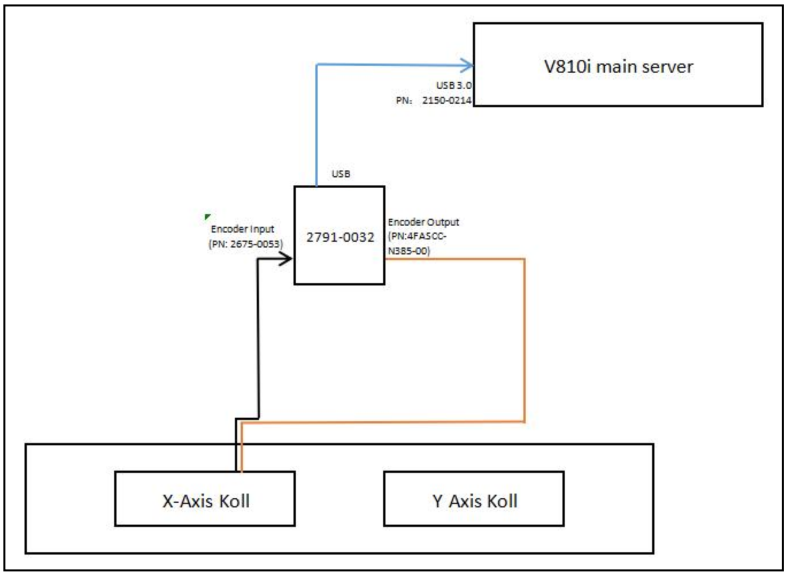


Figure 9: Schematic Diagram

11. **Remove the obstacle from read head encoder:** Take away the paper/plastic that block the signal of read head encoder and proceed for calibration.

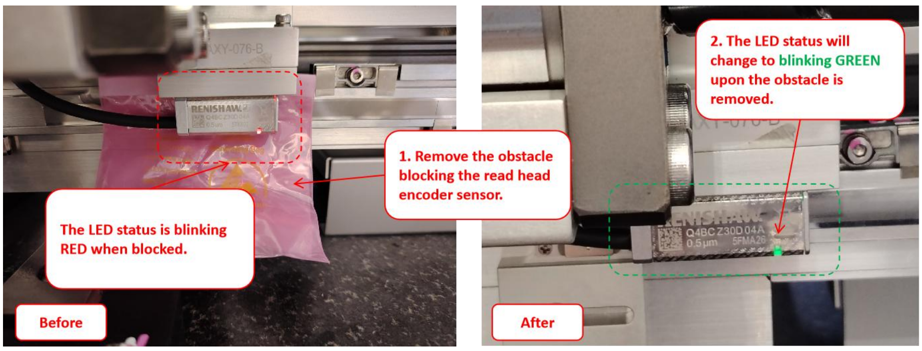


Figure 10: Remove obstacle from Read Head Encoder Sensor

12. **Software Initialization:** Open the ADT view software on the main server and wait for the read head encoder connection to establish. This may take a moment; please be patient.

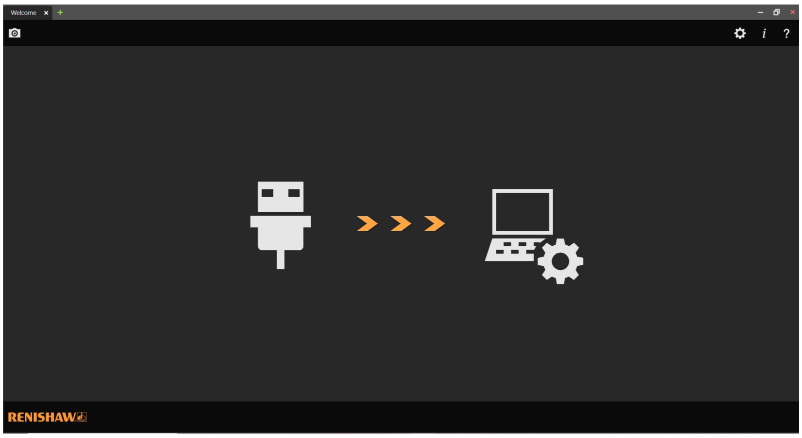


Figure 11: ADT View Startup Page

13. **Connection Confirmation:** Once the connection is established, click on the “Connect” button within the software to proceed with the adjustment and confirmation process.

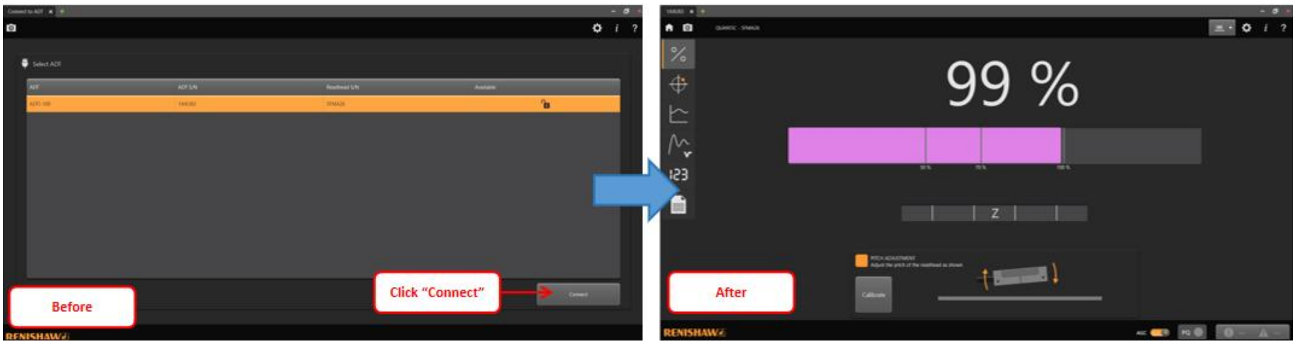


Figure 12: Read Head Encoder Connection with ADT View Application

14. **Ensure AGC is Disabled:** The AGC must be in disabled mode. If not, manually turn it off.

Figure 13: AGC in Disabled Mode

15. **Navigation to Graph Tab:** Navigate to the “Graph” tab within the ADT view software interface to access the stroke analysis tools.

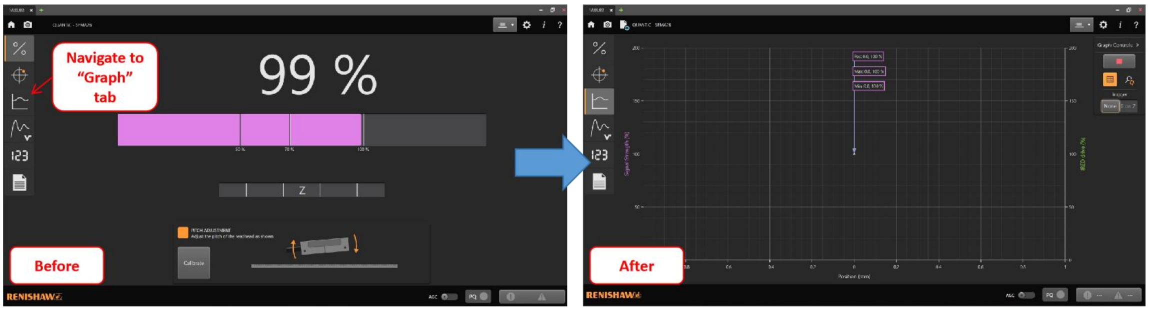


Figure 14: Navigate to “Waveform” Tab

16. **Stroke Analysis:** Manually push the stage from the minimum hard stop (located at the front access door) to the maximum hard stop (located at the rear access door). Analyze the stroke percentage displayed on the software interface. The stroke accuracy percentage should ideally fall within the range of **90%-105%.** If it falls outside this range, proceed to the next step for adjustment.

*Note: “**Pos**” reflects the **current stage position** in live mode, while “**Max**” and “**Min**” reflect the positions with the **highest** and **lowest readings** along the entire stroke, respectively.*

Figure 15: X-Axis Stroke Accuracy Result (Sample Image)

17. **Encoder Adjustment Initiation:** Loosen the two screws securing the read head encoder bracket to initiate the adjustment process.

18. **Adjustment Procedure:** Using precision tools, adjust the position of the read head encoder to bring the stroke percentage accuracy within the desired range of 90%-105%.

Note: For fine tuning, loosen either screw for adjustment to avoid significant fluctuations.

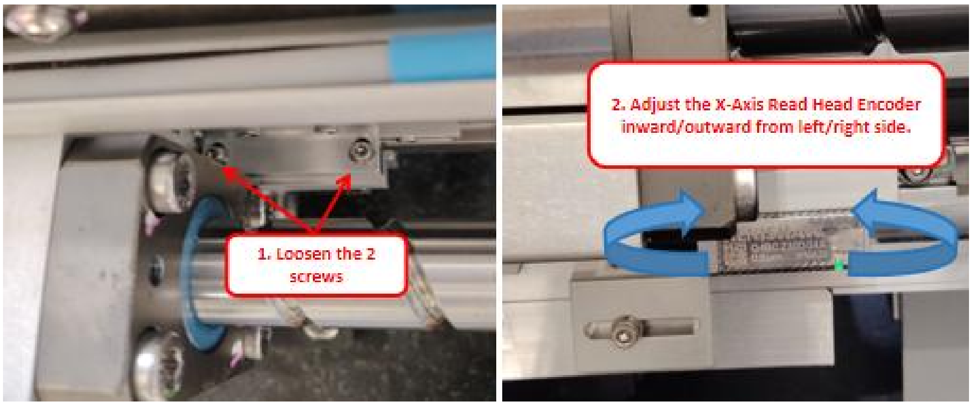


Figure 16: Read Head Encoder Adjustment

19. **Confirmation of Stroke Percentage:** Once the adjustment is made, confirm that the stroke percentage accuracy falls within the specified range by re-analyzing the stroke using the software interface.

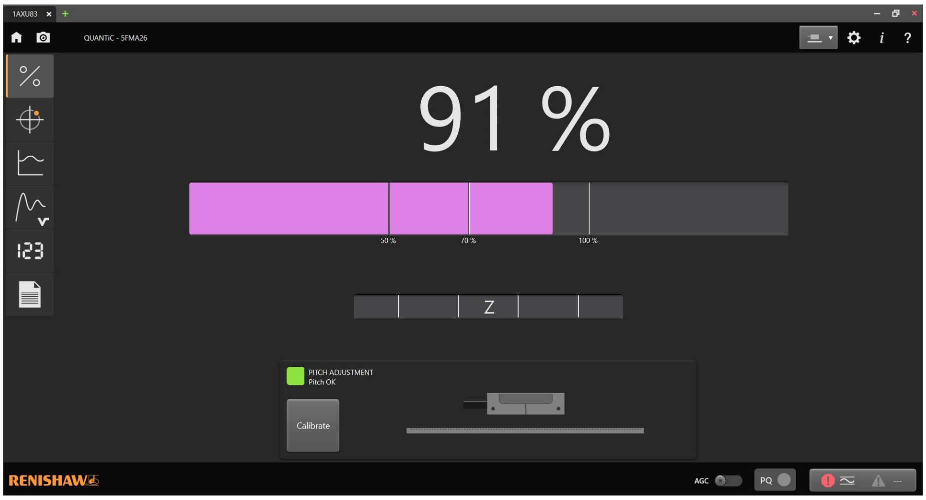


Figure 17: Stroke accuracy percentage (Sample Image)

20. **Bracket Securing:** Upon achieving the desired stroke accuracy, tighten back the two screws for the read head encoder bracket to secure its position.

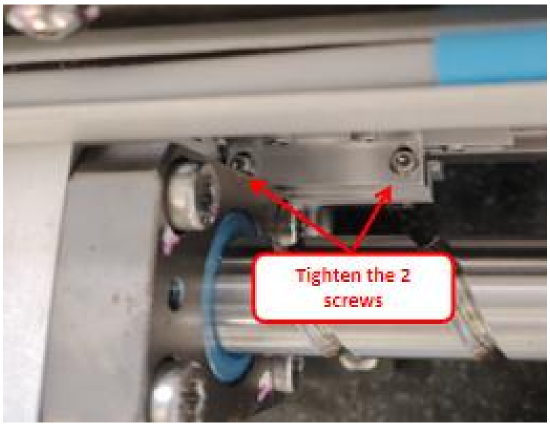


Figure 18: X-Axis Read Head Encoder

21. **Stroke Validation:** Repeat Steps 15 to 16 to ensure that the entire X-Axis stroke falls within the range of 90%-105% signal strength. If out of range, repeat Steps 17-20 for re-adjustment.
22. **Navigation to Calibration Tab:** Navigate to the “%” tab within the software interface and click on the “Calibrate” button to initiate the calibration process.

Figure 19: Read Head Encoder Calibration

23. **Calibration Process:** Follow the on-screen instructions to manually move the stage back and forth without passing through the home sensor. This process helps calibrate the reference mark for accurate positioning.

Figure 20: Incremental Signal Calibration (Sample Image)

24. **Calibration Confirmation:** Once the reference mark calibration message appears, move the stage back and forth to pass through the home sensor. Observe the Z signal bar, which will show green upon calibration completion.

Note: In case AGC is not turned on, manually turn on the Automatic Gain Control (AGC) as instructed.

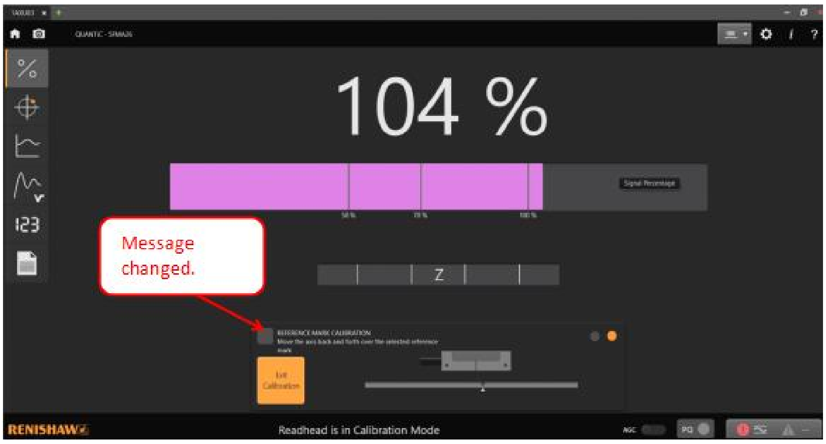


Figure 21: Reference Mark Calibration (Sample Image)

Figure 22: Z signal bar with Green Status (Sample Image)

25. **Connector Handling:** Unplug the connector for the advanced diagnostic tool and reconnect the read head encoder connector according to the manufacturer's instructions.
26. **Machine Performance Verification**: Proceed to section "**Verfication of Machine Performance after Adjustment and Confirmation**" before release the machine to customer.

Y-Axis Encoder Head Alignment Procedure

27. **Read Head Spacer Preparation:** Obtain a read head spacer, ensuring it matches the required specifications outlined by the manufacturer for the alignment process.

    Note: the spacer was attached with the carton box of spare part 2675-0053

Figure 23: Read Head Encoder Spacer

28. **Y-Axis Read Encoder Bracket Loosening:** Using the appropriate tool, carefully loosen the two screws located at the X-Axis read head encoder bracket. Take care not to over-loosen the screws to prevent any accidental detachment of components.

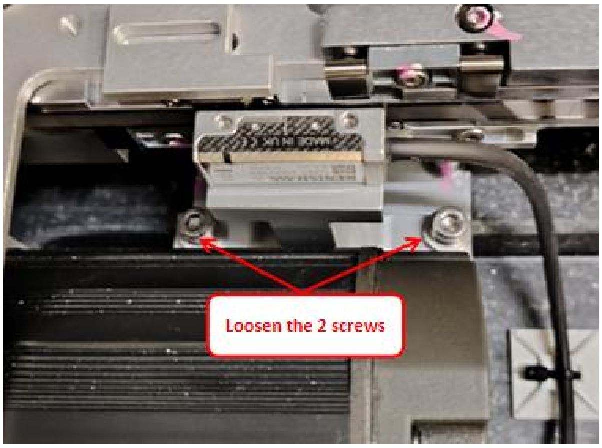


Figure 24: Y-Axis Read Head Encoder

29. **Read Head Alignment Spacer Placement:** Place the read head alignment spacer between the read head encoder and the Quantic scale, ensuring it is positioned centrally and evenly aligned.

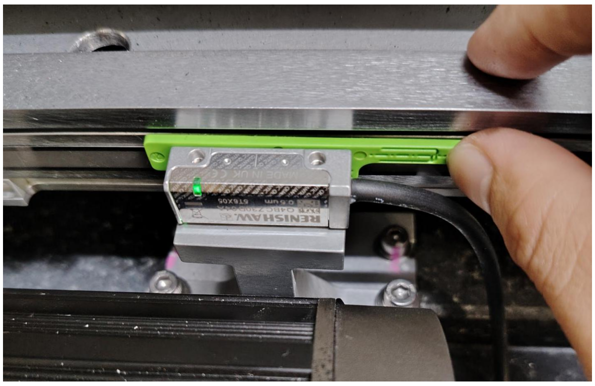


Figure 25: Y-Axis Read Head Encoder with Spacer

30. **Read Head Encoder Adjustment:** Gently push the Read Head Encoder towards the read head spacer until it makes contact. Ensure that the spacer is fully engaged and seated properly. Once in position, use the appropriate tool to tighten the two screws at the X-Axis read head encoder bracket securely. Apply torque evenly to prevent any misalignment or slippage during operation. Remove the spacer after securing the encoder bracket.

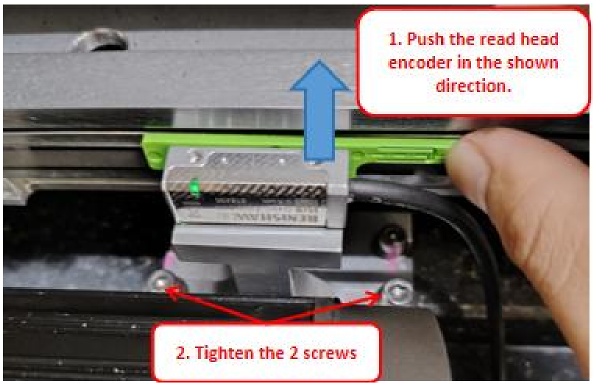


Figure 26: Y-Axis Read Head Encoder Alignment

31. **Stage Movement Verification:** Manually push the stage from the minimum hard stop (located at the front access door) to the maximum hard stop (located at the rear access door). Throughout the movement, observe the rear head encoder LED, which should remain blinking green continuously. This indicates proper alignment and consistent signal detection as the stage moves along the axis.
Note: If the read head encoder LED turns RED/ORANGE at any location of the stage, it indicates potential misalignment. In such cases, route back to Step 26 for read head encoder alignment.
32. **SSCA Power Cycle:** Power off the SSCA (Servo System Control Amplifier) to reset the system. Follow the manufacturer's recommended procedure for safely powering down the SSCA unit.
33. **LED Status Change:** After powering on the SSCA, observe the change in status of the LED on the read head encoder. The blinking green LED will transition to blinking blue, indicating a change in operational mode.

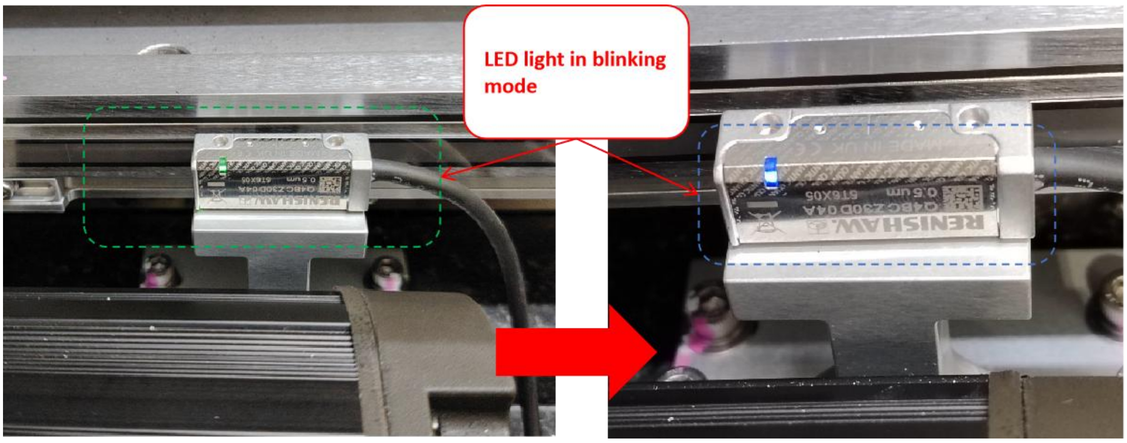


Figure 27: LED Status Change (Blinking Green to Blinking Blue)

34. **Y Stroke Verification:** Using controlled force, push and pull the Y stroke to execute movement along the axis. Ensure that the read head encoder passes through the home sensor magnet once per movement for both forward and backward directions. As the alignment progresses, the blinking blue LED will stabilize and appear as a static blue, indicating successful alignment completion.

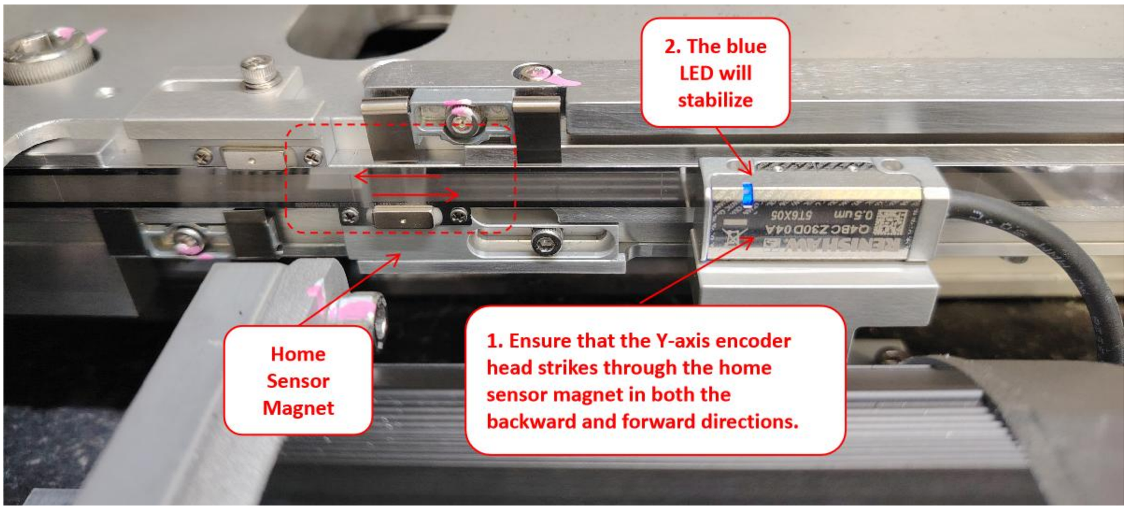


Figure 28: To Stabilize the blue LED for Read Head Encoder (Sample Image)

35. **Block Read Head Encoder Sensor**: Use a piece of paper/plastic to block the encoder head sensors for signal reset before proceed to adjustment and confirmation procedure.

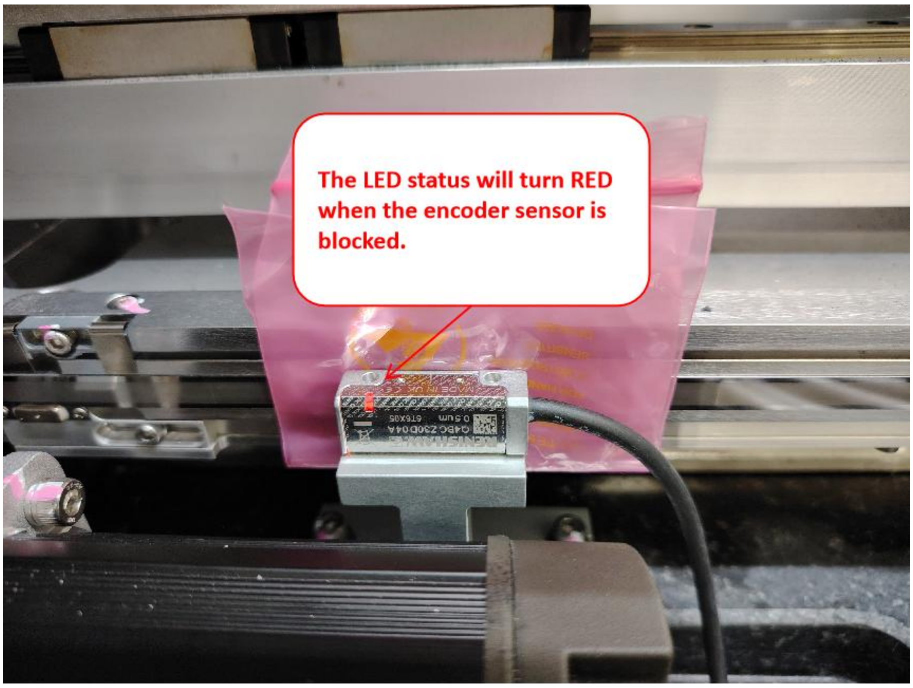


Figure 29: To Reset the LED Status for Read Head Encoder into Detection Mode

Y-Axis Encoder Head Adjustment and Confirmation

36. **Connection Establishment:** Establish connections for the Advanced Diagnostic Tool (ADT), main server, encoder, and Kollmorgen kit according to the manufacturer's instructions.

Note: Be ready with 4XASPP-A440 (Renishaw Quantic Encoder Alignment Kit)

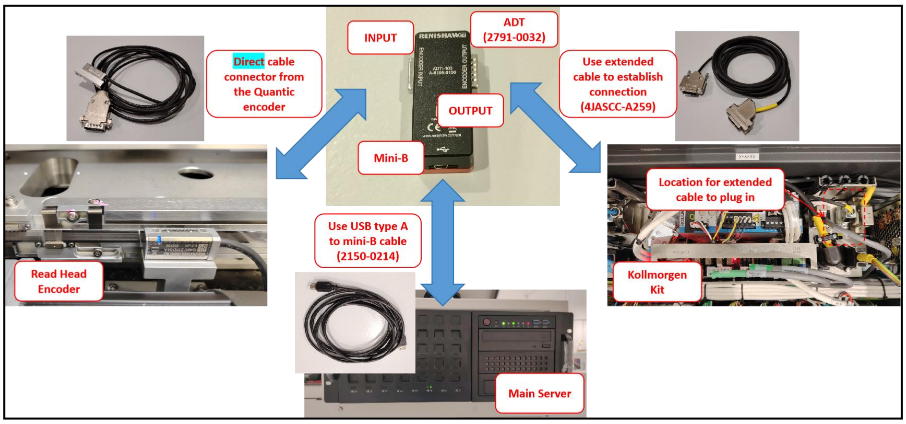


Figure 30: ADT, Main Server, Encoder and Kollmorgen Kit Connections

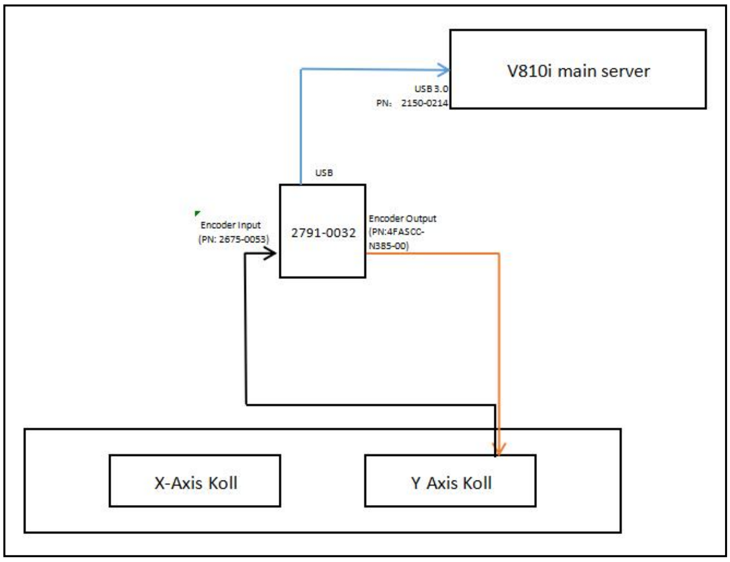


Figure 31: Schemetic Diagram

37. **Remove the Obstacle from Read Head Encoder:** Take away the paper/plastic that blocking the signal of read head encoder and proceed for calibration.

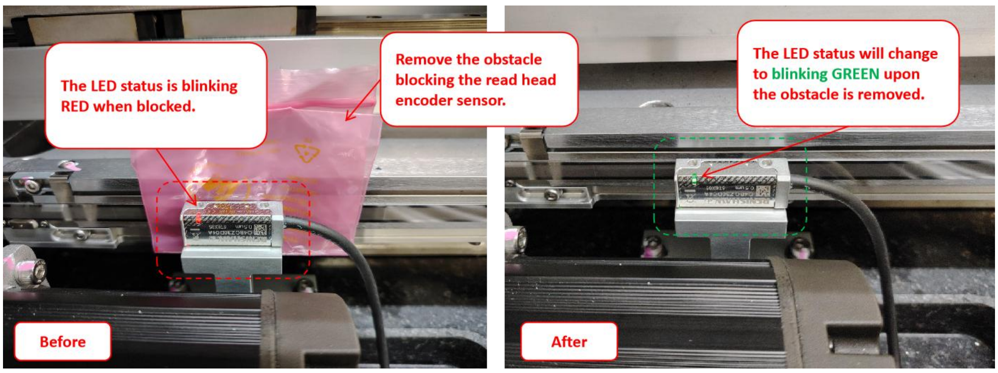


Figure 32: Remove obstacle from Read Head Encoder Sensor

38. **Software Initialization:** Open the ADT view software on the main server and wait for the read head encoder connection to establish. This may take a moment; please be patient.

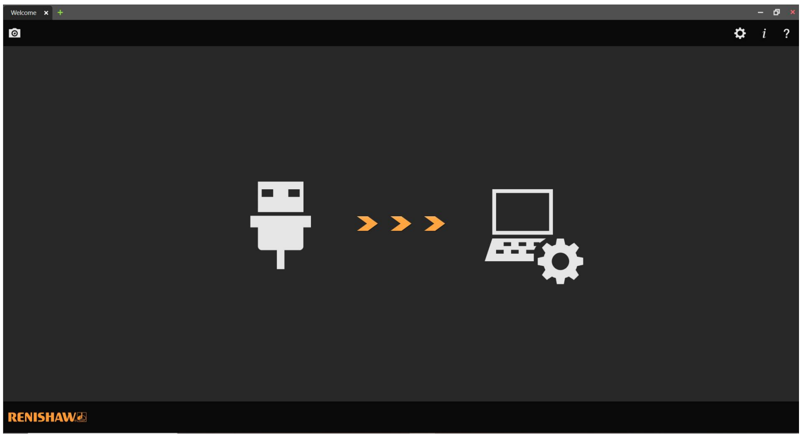


Figure 33: ADT View Startup Page

39. **Connection Confirmation:** Once the connection is established, click on the "Connect" button within the software to proceed with the adjustment and confirmation process.

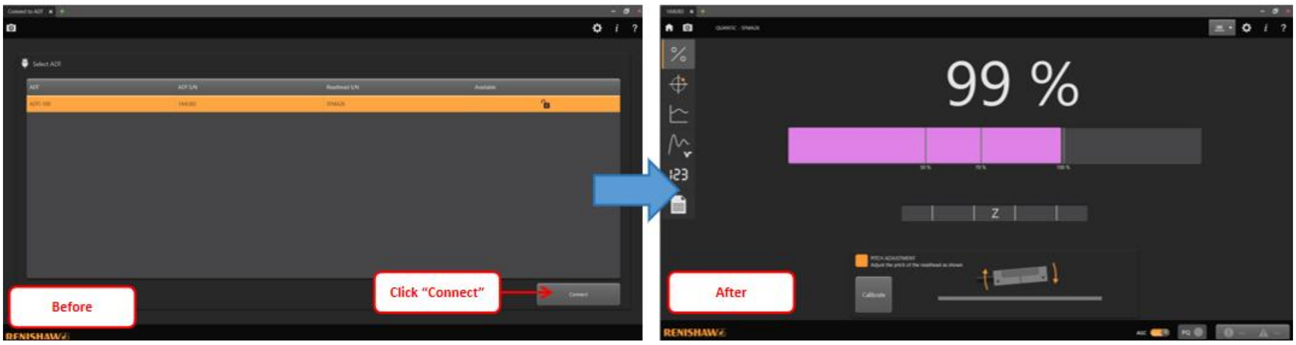


Figure 34: Read Head Encoder Connection with ADT View Application (Sample Image)

40. **Ensure AGC is Disabled:** The AGC must be in disabled mode. If not, manually turn it off.

Figure 35: AGC in Disabled Mode

41. **Navigation to Graph Tab:** Navigate to the “Graph” tab within the ADT view software interface to access the stroke analysis tools.

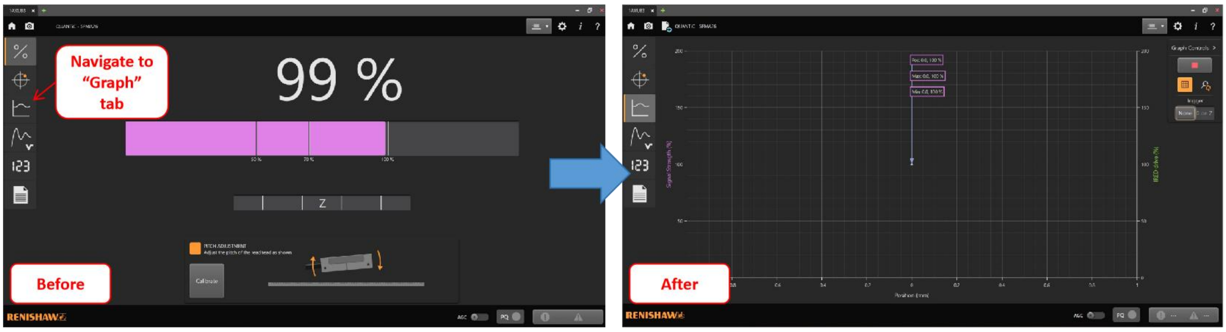


Figure 36: Navigate to “Waveform” Tab

42. **Stroke Analysis:** Manually push the stage from the minimum hard stop (located at the front access door) to the maximum hard stop (located at the rear access door). Analyze the stroke percentage displayed on the software interface. The stroke accuracy percentage should ideally fall within the range of **90%-105%.** If it falls outside this range, proceed to the next step for adjustment.

*Note: “**Pos**” reflects the **current stage position** in live mode, while “**Max**” and “**Min**” reflect the positions with the **highest** and **lowest readings** along the entire stroke, respectively.*

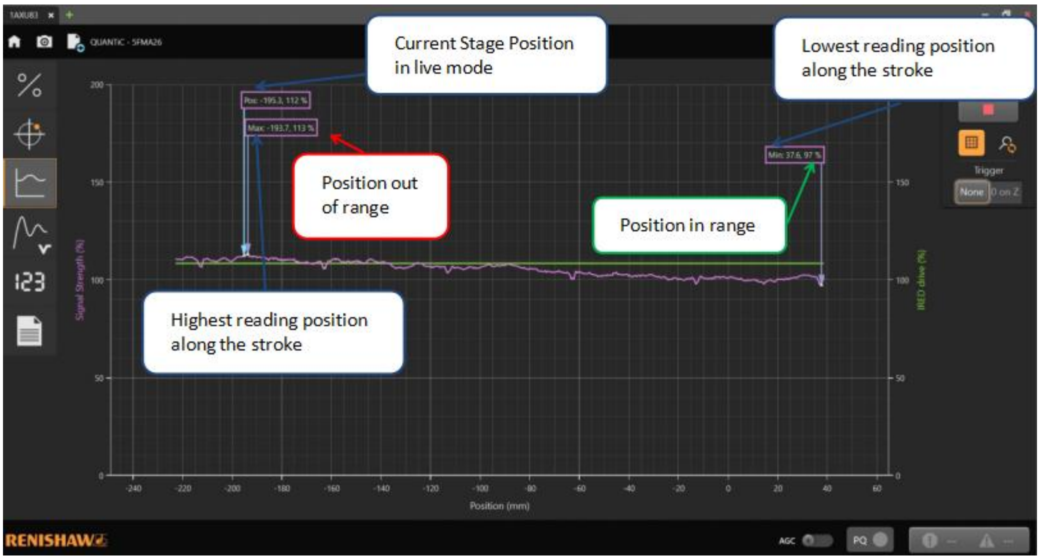


Figure 37: Y-Axis Stroke Accuracy Result (Sample Image)

43. **Encoder Adjustment Initiation:** Loosen the two screws securing the read head encoder bracket to initiate the adjustment process.

44. **Adjustment Procedure:** Using precision tools, adjust the position of the read head encoder to bring the stroke percentage accuracy within the desired range of 90%-105%.

Note: For fine tuning, loosen either screw for adjustment to avoid significant fluctuations.

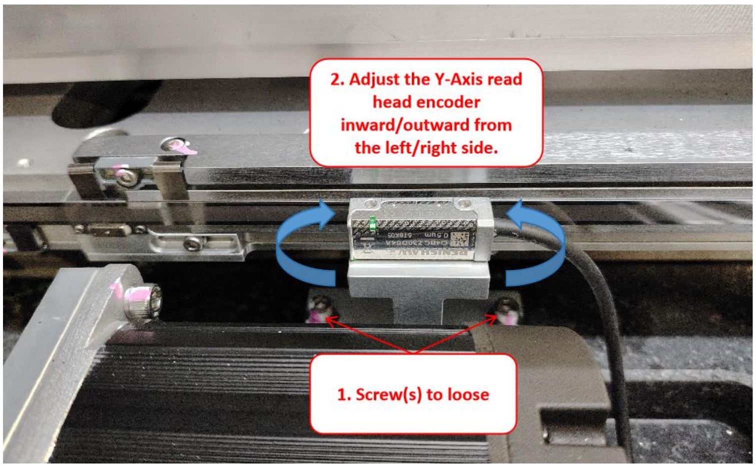


Figure 38: Read Head Encoder Adjustment

45. **Confirmation of Stroke Percentage:** Once the adjustment is made, confirm that the stroke percentage accuracy falls within the specified range by re-analyzing the stroke using the software interface.

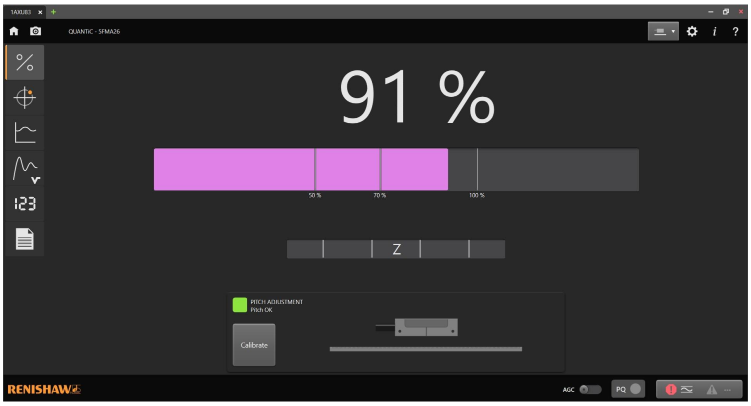


Figure 39: Stroke accuracy percentage (Sample Image)

46. **Bracket Securing:** Upon achieving the desired stroke accuracy, tighten back the two screws for the read head encoder bracket to secure its position.

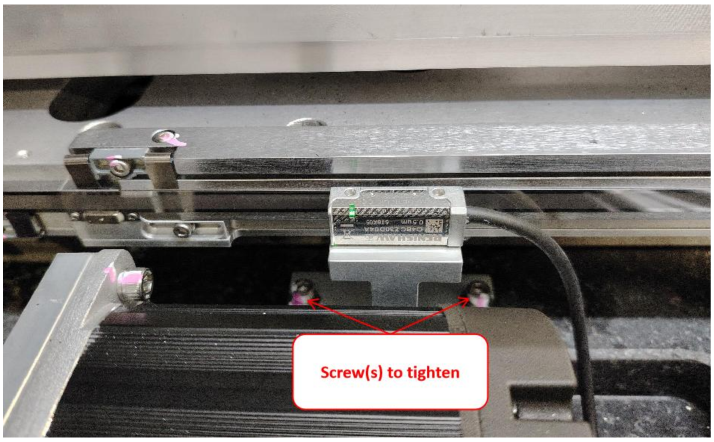


Figure 40: Y-Axis Read Head Encoder

47. **Stroke Validation:** Repeat Steps 40 to 41 to ensure that the entire Y-Axis stroke falls within the range of 90%-105% signal strength. If out of range, repeat Steps 42 to 45 for re-adjustment.
48. **Navigation to Calibration Tab:** Navigate to the "%" tab within the software interface and click on the"Calibrate"button to initiate the calibration process.

Figure 41: Read Head Encoder Calibration

49. **Calibration Process:** Follow the on-screen instructions to manually move the stage back and forth without passing through the home sensor. This process helps calibrate the reference mark for accurate positioning.

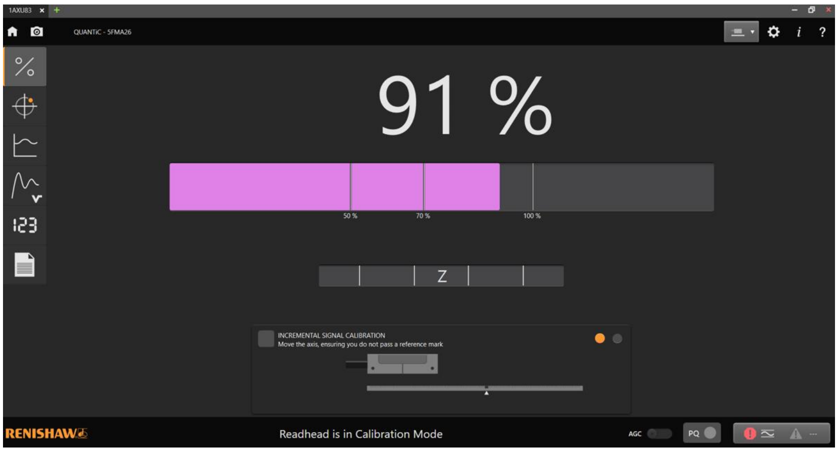


Figure 42: Incremental Signal Calibration (Sample Image)

50. **Calibration Confirmation:** Once the reference mark calibration message appears, move the stage back and forth to pass through the home sensor. Observe the Z signal bar, which will show green upon calibration completion. Manually turn on the Automatic Gain Control (AGC) as instructed.

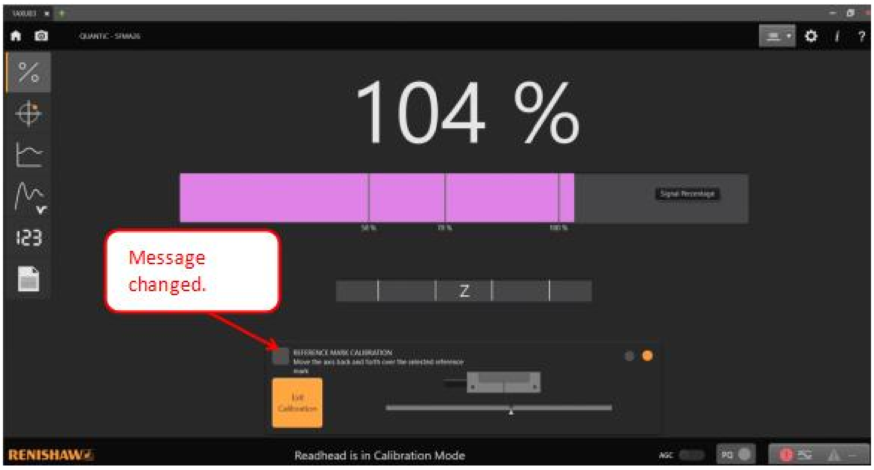


Figure 43: Reference Mark Calibration (Sample Image)

Figure 44: Z signal bar with Green Status (Sample Image)

51. **Connector Handling:** Unplug the connector for the advanced diagnostic tool and reconnect the read head encoder connector according to the manufacturer's instructions.
52. **Machine Performance Verification**: Proceed to section "**Verfication of Machine Performance after Adjustment and Confirmation**" before release the machine to customer.

Validation of Machine Performance After Adjustment and Confirmation

53. **Tool Removal:** After completing the alignment process, ensure all tools, equipment, and debris are removed from the machine stage and surrounding area. Collect any tools used during the adjustment and confirmation procedures and return them to their designated storage locations.
54. **Access Door Switching:** Proceed to each access door of the machine and switch them back to production mode. Ensure that each access door is securely closed and locked according to standard operating procedures and verify that all safety interlocks associated with the access doors are engaged and functioning properly.
55. **Subsystem Initialization:** Power on the machine and run V810 software. Perform a login procedure to access the system interface. Navigate to the 'Services > Subsystem Status & Control > Panel Positioner' section within the software interface. Click on the "Initialize" button to reset the panel positioner subsystem to its default settings. Wait for the initialization process to complete before proceeding further.
56. **Long Scan Execution:** Execute five loops of long scan procedures using the machine's scanning capabilities. Ensure that the scan parameters are set according to the desired specifications and imaging requirements. Monitor the scan progress and observe the performance of the machine throughout the scanning process.
57. **CD&A Loop Execution:** Run one loop of CD&A (Comparison, Dosage, and Alignment) procedures to further validate the performance of the machine. Follow the standard operating procedures for CD&A loop execution, ensuring that all necessary parameters and settings are configured correctly.
58. **Monitoring and Verification of Machine Performance:** Monitor the execution of CD&A procedures and carefully observe the machine's performance, including accuracy, stability, and consistency in delivering results. Verify that the machine operates within acceptable tolerances and meets the required performance standards.